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School of Science Engineering and Technology Skills

Department of Engineering

Subject – AWS Lab

Experiment – 9

Title: AWS Billing, Cost Management, and Monitoring

1. Introduction

Managing costs is a critical pillar of the **AWS Well-Architected Framework**. As cloud environments scale, understanding the financial impact of infrastructure choices becomes essential. This lab covers the tools required to estimate future costs, set spending limits, and analyze historical usage trends to prevent "bill shock."

2. Lab Objectives

1. Estimate the monthly cost of a web architecture using the **AWS Pricing Calculator**.
2. Configure an **AWS Budget** to track actual vs. forecasted spending.
3. Set up a **Billing Alarm** in CloudWatch to trigger notifications at specific price points.
4. Utilize **AWS Cost Explorer** to visualize and identify patterns in resource consumption.

3. Prerequisites

- An active AWS Management Console account.
- Basic understanding of AWS services (EC2, S3, RDS).
- Permissions to access the Billing and Cost Management Dashboard.

Part 1: Estimating Costs with AWS Pricing Calculator

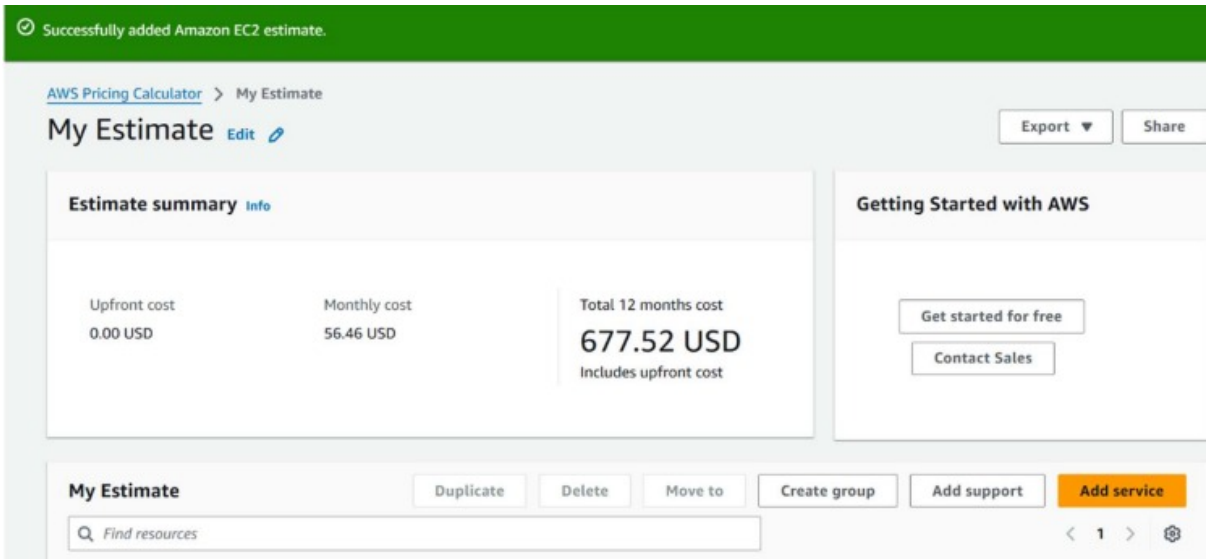
Before deploying resources, architects use this tool to model costs based on specific configurations.

Step-by-Step Instructions:

1. Open the AWS Pricing Calculator.
2. Click **Create estimate**.
3. **Search for Service:** Type EC2 and click **Configure**.
4. **Define Strategy:**

- **Region:** Select your closest region (e.g., US East-1).
- **Instance Type:** Search for t3.medium.
- **Quantity:** Set to 2.

5. **Add Storage:** Scroll to Amazon EBS and configure a 20 GB General Purpose SSD (gp3) volume.
6. Click **Add to estimate** and then **View summary** to see the total monthly and annual projected cost.

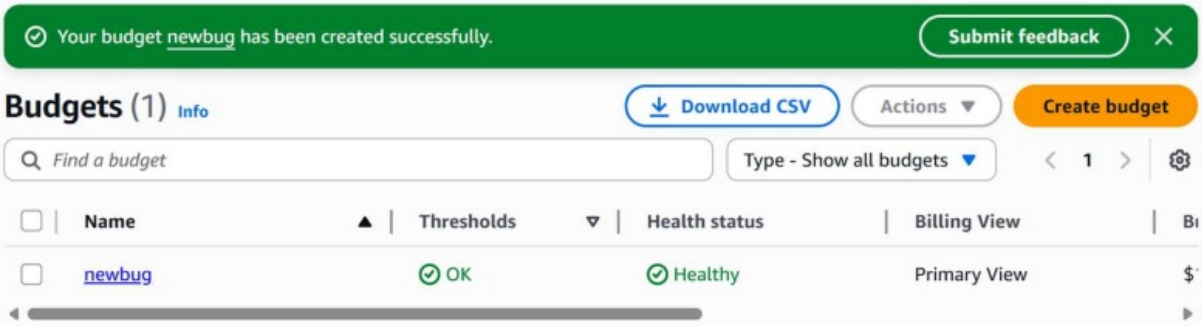


Part 2: Tracking Costs with AWS Budgets

Budgets allow you to stay informed about your spend and can even trigger automated actions.

Step-by-Step Instructions:

1. In the AWS Console, search for **Billing and Cost Management**.
2. In the left sidebar, select **Budgets** > **Create budget**.
3. **Budget Type:** Choose **Cost budget (Recommended)**.
4. **Set Amount:**
   - **Name:** Monthly-Cloud-Limit.
   - **Period:** Monthly.
   - **Budgeted Amount:** Enter a small value for lab purposes (e.g., $10.00).
5. **Configure Alerts:**
   - Set an alert for **Actual** costs at **80%** of the budget.
   - Enter your email address for the notification.
6. Click **Create budget**.

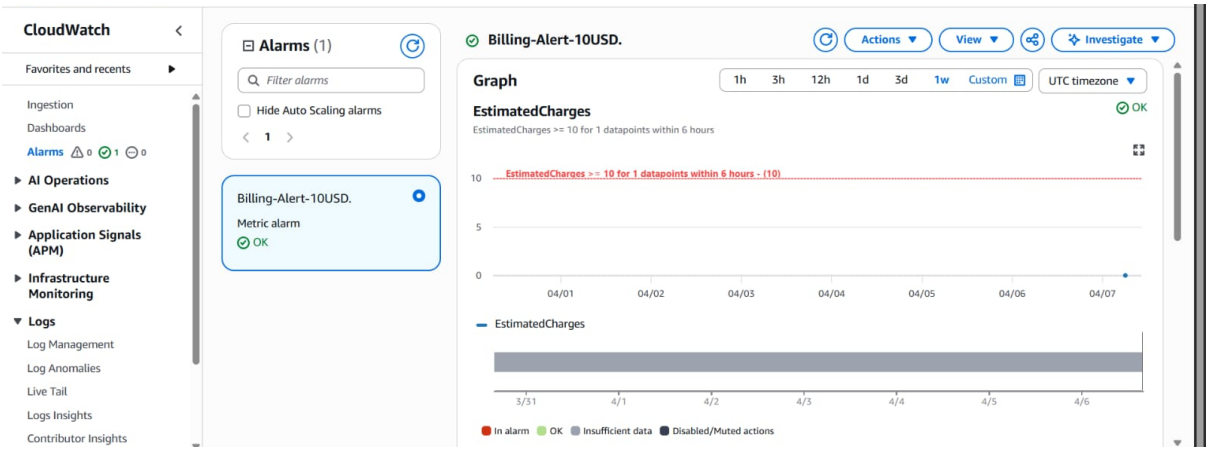


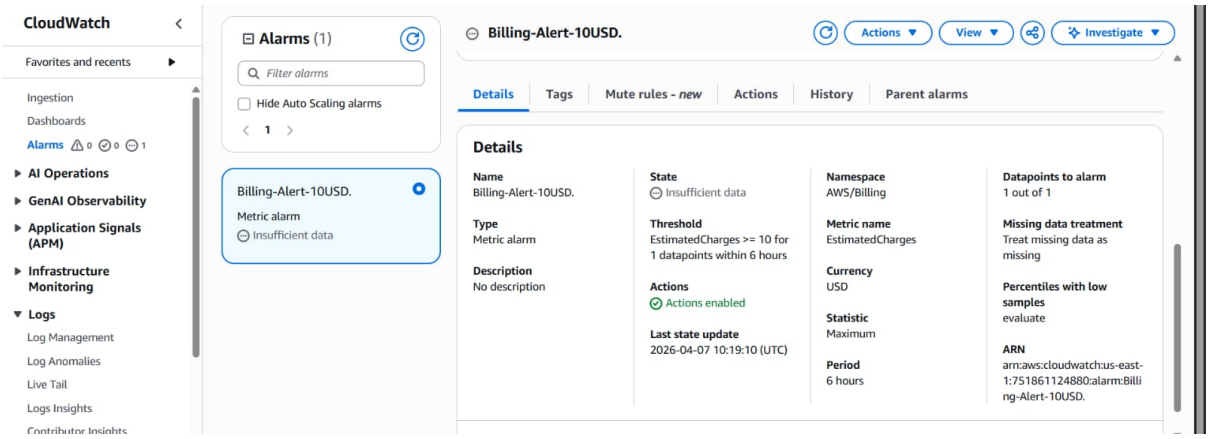
Part 3: Setting Up Billing Alarms in CloudWatch

While Budgets are for high-level oversight, CloudWatch Billing Alarms provide a direct metric-based trigger.

Step-by-Step Instructions:

1. **Note:** Billing metrics are stored in the **US East (N. Virginia)** region regardless of where your resources are. Switch your region to us-east-1.
2. Go to the **CloudWatch** console.
3. In the sidebar, select **Alarms** > **All alarms** > **Create alarm**.
4. Click **Select metric** > **Billing** > **Total Estimated Charge**.
5. Check the box for the EstimatedCharges metric and click **Select metric**.
6. **Conditions:** Set the threshold to **Static** > **Greater than** > $5.
7. **Notification:** Select an existing SNS topic or create a new one to email you when the charge exceeds $5.
8. Save the alarm as Billing-Alert-5USD.



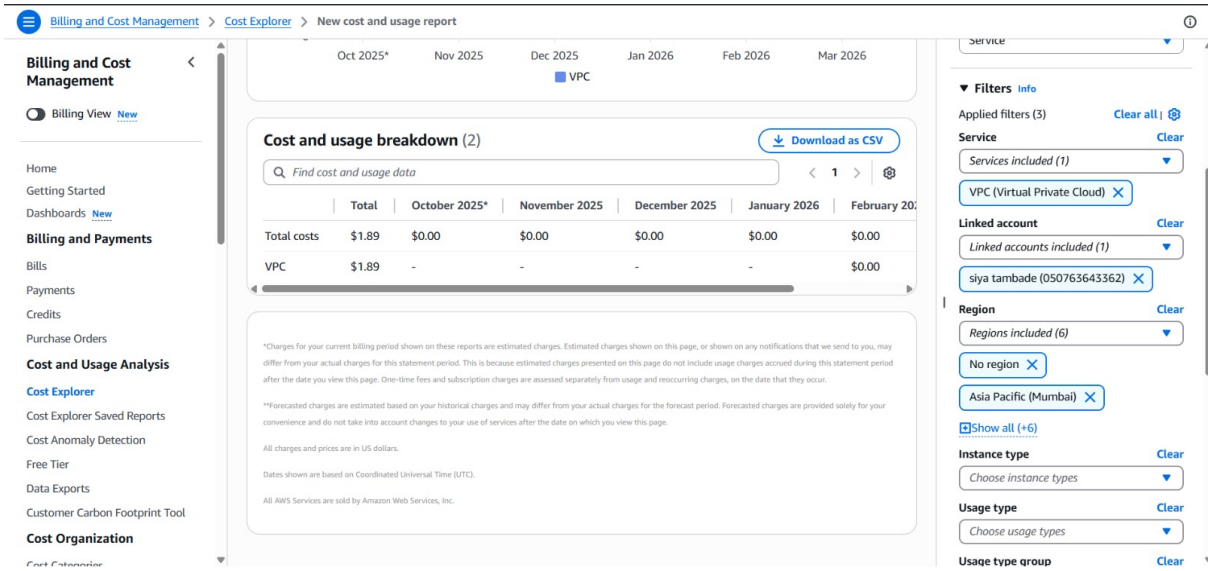


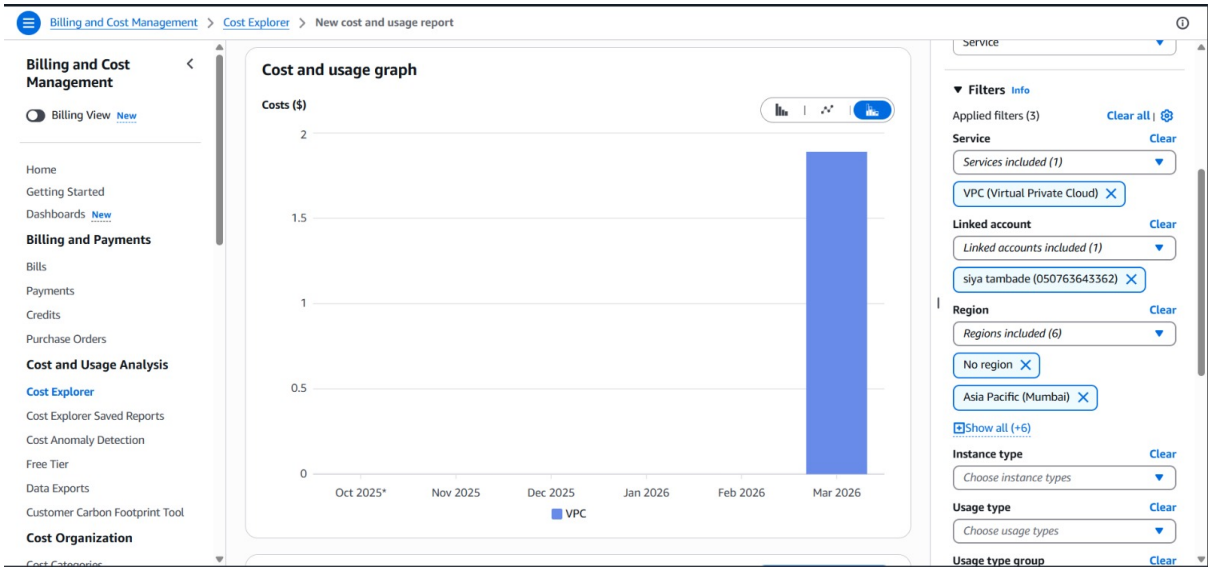
Part 4: Analyzing Trends with Cost Explorer

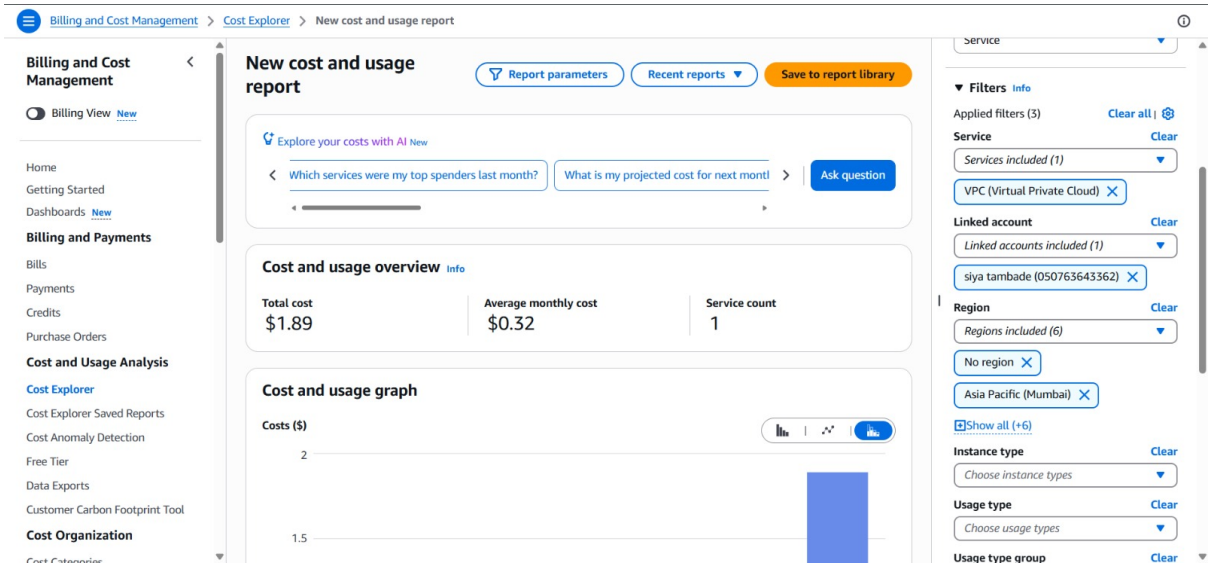
Cost Explorer helps you see "where the money went" by breaking down costs by service or tag.

Step-by-Step Instructions:

1. Navigate back to the **Billing and Cost Management** dashboard.
2. In the sidebar, select **Cost Explorer**.
3. **Launch Cost Explorer:** If this is your first time, it may take up to 24 hours to populate data.
4. **Filter Data:**
   - **Service:** Filter by Amazon EC2.
   - **Group by:** Select Usage Type or Region.
5. **Identify Trends:** Look for spikes in the bar chart and identify which day or service caused the increase.
6. **Save Report:** Click **Save as** and name it EC2-Monthly-Trend.







Observation Table

Tool	Purpose	Key Observation (Example)
Pricing Calculator	Cost Estimation	2 x t3.medium = ~$XX.XX/month
AWS Budgets	Spending Guardrails	Alert set at 80% of $10 budget
CloudWatch Alarm	Metric Thresholds	Triggered when total charges > $5
Cost Explorer	Historical Analysis	EC2 is 60% of total monthly spend

Conclusion

Cost management is an ongoing process. Through this lab, you have learned to estimate costs before deployment, set up proactive alerts to prevent overspending, and perform deep-dive analysis of usage patterns. This ensures your cloud environment remains cost-optimized and financially predictable.